Table of Trial Solutions for the Method of Undetermined Coefficients

We can choose an *Ansatz* to the particular solution based on the following table:

	$f(t)$	y_p
Constant	C	A_0
nth order polynomial	$P_n(t) = P_n t^n + P_{n-1}t^{n-1} + \cdots + P_0$	$A_n(t) = A_n t^n + A_{n-1}t^{n-1} + \cdots + A_0$
Exponential	$e^{\alpha t}$	$A_0 e^{\alpha t}$
Trigonometric	$\cos(\beta t)$, $\sin(\beta t)$, or $\cos(\beta t) + \sin(\beta t)$	$A_0 \cos(\beta t) + B_0 \sin(\beta t)$
Polynomial × Exponential	$P_n(t)e^{\alpha t}$	$A_n(t)e^{\alpha t}$
Polynomial × Trigonometric	$P_n(t)\cos(\beta t) + Q_n(t)\sin(\beta t)$	$A_n(t)\cos(\beta t) + B_n(t)\sin(\beta t)$
Exponential × Trigonometric	$e^{\alpha t}\cos(\beta t) + e^{\alpha t}\sin(\beta t)$	$A_0 e^{\alpha t}\cos(\beta t) + B_0 e^{\alpha t}\sin(\beta t)$

Note: If some terms of y_p are found in $y_h(t)$ (i.e. such terms are the same as the homogeneous solution), multiply the expression for $y_p(t)$ by t (or, if necessary, by t^2) to eliminate the duplication.